

EDUCATION

2022-2026

B.E In Mechanical Engineering
American University of Beirut

- Proficient in mechanical design tools, with expertise in green energy technologies, sustainability practices, HVAC, Thermo and heat transfer, corrosion detection research, 3D modeling and assembly design , reverse engineering, and sustainability and engineering standards , computational fluid dynamics , robotics.
- Analytical and driven Mechanical Engineering student with hands-on experience in data analysis, business strategy, and project delivery. Proven ability to close deals, analyze markets, and drive key marketing decisions in entrepreneurial ventures.
- Co-founder and events officer cabinet of AEE AUB chapter.

SKILLS & ABILITIES

Excel	Computer Skills	Customer Service	LabView	Robotics	PCBs
Ansys Fluent/Thermal	Reports/Word	MATLAB/Simulink	AutoCAD	SolidWorks	ML for data analysis

EXPERIENCE

Jun 2023-Jan 2024

Marketing Associate Intern, TravelnAi
Implemented Marketing, Sales, and customer service in an app for traveling suggestions with the help of AI.

- Conducted market research and competitor analysis to identify strategic positioning for AI-based travel app.
- Closed B2B deals by presenting data-driven value propositions to clients.

May-June 2025

Automotive Intern, Porsche
Assisted in the restoration and maintenance of Porsche vehicles with hands-on and theoretical exposure to mechanical and electrical systems.

- Helped rebuild a classic Porsche racing car, focusing on material properties and fault diagnostics.
- Gained hands-on experience with EV and ICE systems, learning key components and repair procedures.
- Learned the full vehicle management cycle—from intake to delivery to the customer.

Sep 2023-Present

Salesman, Self Employed
Applied project management and problem-solving skills to deliver customized technical solutions and services for local businesses and an AUB community manager (student based)

July -Sept 2025

Automotive Intern, RYMCO (Rasamny Younes Motor Co)

- Collaborated with technicians on Nissan, Infiniti, GMC, McLaren, including Electric and Hybrid vehicles.
- Served as an engineering manager by supervising technicians and evaluating their performance during Nissan’s technical competition.
- Analyzed engineering design, workflow, material selection, lifetime performance, and heat transfer.
- Studied and assessed car HVAC systems, efficiency optimization, control systems, and sensor performance.
- Prepared and reviewed engineering reports for submission to the head company and management.
- Evaluated engine performance, manufacturing plans, and check-up schedules.
- Conducted Pre-Delivery Inspections (PDI) to ensure compliance and quality standards.
- Participated in test and practice days, discussing design, flaws, and safety measures.
- Supported recall procedures to address engineering issues and enhance safety.

PROJECTS & RESEARCH EXPERIENCE

- **13-month Research course concerning hyperspectral imaging with Dr.Samir Mustafa-AUB**
Topics: Accelerated Corrosion - Corrosion in concrete-Corrosion under paint -Corrosion Detection using ML over HSI images
 - Performed advanced data analysis on reflectance spectra using Python and MATLAB, applying continuum removal and signal preprocessing techniques.
 - Brainstormed Ideas and testing procedures and was responsible for the experimental setup and its maintenance.
 - Delivered a comprehensive final presentation with visual and technical reports.
- **Industrial project** with precise calculations for an economical solution for 24-hour electricity availability comparing generators and solar panels calculating efficiency, price, and lifetime factors for the final present worth analysis and breakeven point.
 - Developed Excel-based present worth analysis comparing solar panel vs generator energy solutions under variable loads and life cycle costs.
- Gained insight into sustainability laws and engineering approaches while exploring innovative green technologies through specialized masterclasses.(AEE, internship)
- **Introduction to Machine Learning — AUB**
Hands-on ML in Python: regression, SVMs, neural nets, transformers, and ethical AI.([github:Raneem1049](#))
- Conducted Computational Fluid Dynamics (CFD) optimization of a vertical-axis wind turbine as part of the Final Year Project, focusing on performance enhancement and aerodynamic efficiency.
- Designed and implemented a Lag-Lead control system to achieve low overshoot and stable ball-height regulation in an Arduino-based fan setup.
- 2nd place in D2T which featured creating a new app that uses AI for a Sustainability-related Invention.
- 4th place in Amazon industry program where we got a lot of knowledge concerning business plans and how to design, market and make money from the app.